Curriculum Vitae

Kurniawan Teguh Waskito, Ph.D.

Department of Mechanical Engineering | Tropical Renewable Energy Center
Faculty of Engineering, Universitas Indonesia | UI Campus, Depok 16426, Indonesia
waskito@eng.ui.ac.id | kurniawan.teguh03@ui.ac.id

PERSONAL DETAILS		
	Nationality	Indonesia
	Research Interests	Marine Hydrodynamics, Ship Performance in Actual Seas, Ocean Renewable Energy, Hydrogen Energy, Wind Energy, Floating Solar PV
	Profile Pages	Scopus Profile ORCID: 0000-0002-9719-9559 Google Scholar Dept. of Mechanical Engineering – Faculty Page

EDUCATION	
2009–2013	**Bachelor of Engineering, Naval Architecture** Universitas Indonesia, Indonesia **Undergraduate thesis:** *Experimental Study and CFD Prediction of Total Hull Resistance of an Asymmetric Pentamaran Model with Separation and Staggered Arrangement Variations of the Inner Side Hulls*
2012–2014	**Master of Engineering (Fast-track), Mechanical Engineering** Universitas Indonesia, Indonesia **Master's thesis:** *Drag Reduction of Crude Oil Flow with CMC Additive and Bacterial Cellulose Fibre Suspension in Circular and Spiral Pipes*
2017–2020	**Doctor of Engineering, Naval Architecture & Ocean Engineering** Osaka University, Japan **Dissertation:** *Prediction of Vertical Bending Moment Using Measured and Computed Wave-Induced Pressure Distribution on Ship Hull*

PROFESSIONAL EXPERIENCE	
2016–Present	**Lecturer** Department of Mechanical Engineering, Universitas Indonesia
2016–Present	**Industrial Engineering Expert and Consultant** P2M, Department of Mechanical Engineering, Universitas Indonesia *Turbomachinery, Hydraulics & Pneumatics, Reliability Engineering, Machine Health Diagnosis & Prognosis*
2021–2022	**Postdoctoral Fellowship – Associate Professorship** Faculty of Maritime Sciences, Kobe University, Japan
2020–2021	**Postdoctoral Fellowship, Fluids Engineering & Ship Performance Evaluation** National Maritime Research Institute (NMRI), Japan
2014–2016	**Field Operation Professional – Coil Tubing** Halliburton Oil Field Services, Indonesia

	Oil and gas well pressure pumping (workover & well intervention); operation and maintenance of coil tubing, pumps, wellheads, and N2 equipment

INDUSTRIAL PROJECT EXPERIENCE	
2025–2026	**Piping and Mechanical Engineer** Feasibility Study for Development and Operation of a Terminal Automation System – Fuel & LPG Product Distribution, PT. Pertamina Patra Niaga
July 2025	**Industrial Trainer** Maintenance Management, PT. Steel Pipe Industry of Indonesia
June 2025	**Industrial Trainer** Pump Operation, Maintenance, and Troubleshooting, PT. Coca Cola Indonesia
Feb–Apr 2025	**Technical Expert** Survey and Assessment of the Bank Indonesia Representative Office Building (KPWDN) – MEP Systems
2024–2025	**Technical Expert** Expert Technical Documentation, PT. United Tractor
2024–2025	**Technical Expert** Deck Cargo Optimisation for Crew Boats and Supply Vessels, PT Pertamina Hulu Energi OSES
2024–2026	**Technical Expert** Competency Assessment Development, Power Generation, PT. PLN
December 2024	**Industrial Trainer** Pump Operation, Maintenance, and Troubleshooting, PT. Kaltim Parna Industri
2024–2025	**Project Manager** Design & Fabrication of Oil Spill Response Equipment, PT. Pertamina Hulu Energi Offshore South East Sumatra
September 2023	**Industrial Trainer** Maintenance Development, PT. Veolia Services Indonesia
July 2023	**Industrial Trainer** Maintenance Management, PT. Veolia Services Indonesia
June 2023	**Industrial Trainer** Reliability Engineering, PT. Prima Transportasi Servis Indonesia
June 2023	**Industrial Trainer** Maintenance Development & Planning, PT. Indonesia Power
March 2023	**Industrial Trainer** Planning, Scheduling, and Maintenance of Hydraulic System Units, PT. Epiroc Southern Asia
Oct–Dec 2017	**Technical Expert** Process Engineering Competency Mapping, PT. Medco E&P Indonesia
Jan–Oct 2017	**Technical Expert** Turbomachinery Training Module Development, PT. Medco E&P Indonesia
Apr–Aug 2016	**Industrial Trainer** In-house Turbomachinery Training, PT. Pertamina EP Subang
Aug 2014–Feb 2016	**Field Operations Professional** Hydraulic and Pneumatic Systems for Oil and Gas Fields, High-Pressure Plunger Pumps, Preventive and Condition-Based Maintenance

RESEARCH GRANTS	
2026–2028	**Co-Investigator** Sustainable Technology for Electroconversion of Palm-Oil Glycerol into Lactic Acid with Hydrogen Co-Production Using Nickel Alloy Catalysts *Palm Oil Research Grant (GRS), BPDPKS*
2025–2026	**Principal Investigator** Development of Platinum- and Ruthenium-Catalysed Electrolysers for Green Hydrogen Production in Marine Energy Applications *Innovation Grant FTUI, Universitas Indonesia*
2025–2026	**Co-Investigator** Real-time Multi-Objective Optimisation of Oscillating Water Column (OWC) Performance Using Surrogate Models *RIIM Competitive Grant, BRIN*
2025–2026	**Principal Investigator** Floater Design Optimisation Using Hybrid Design of Experiments for a Multi-Cylinder Floating Point Absorber Wave Energy Converter: A Pelabuhanratu Case Study *PUTI Q1, Universitas Indonesia*
2024–2025	**Principal Investigator** CFD Hydrodynamic Analysis of Floater Geometry for a Double-Cylinder Floating Point Absorber Wave Energy Converter (WEC) *Seed Funding (Lector), Faculty of Engineering, Universitas Indonesia*
2024–2025	**Principal Investigator** Nearshore Power Generation Technology (Floating PV, Wind, etc.) *Prime Engineering Seed Fund Batch I 2024*
2024–2025	**Co-Investigator** Estimation of Hydraulic Power Take-Off Parameters for a Floating Absorber Wave Energy Converter Using a Genetic Algorithm *PUTI Q2, Universitas Indonesia*
2024–2025	**Principal Investigator** Seakeeping Performance Prediction of Passenger Ferries in Intact and Damaged Conditions Based on Probabilistic Damage Stability Using CFD and Potential Flow Theory *PUTI Q2, Universitas Indonesia*
2024–2025	**Principal Investigator** Design of a Hydraulic Power Take-Off (HPTO) System with a Double-Cylinder Floating Point Absorber for Coastal Wave Energy Converters *PUTI Q1, Universitas Indonesia*
2022–2025	**Co-Investigator** Electronic Monitoring System for Capture Fisheries (Fish-monitoring) *RISPRO Invitation, LPDP*
2022–2023	**Co-Investigator** Innovative Power Take-Off (PTO) Design Based on Hydraulic Cylinders and Multi-Absorber Arrays for Coastal Wave Energy Converters *PUTI Q2, Universitas Indonesia*
2022–2023	**Co-Investigator** Renewable Power Plant Concept for Wave Energy Conversion: Hydraulic Power Take-Off (PTO) System with Multiple Absorbers *PDUPT, Ministry of Education, Culture, Research and Technology*

BOOK

Yanuar, Gunawan, **Kurniawan T. Waskito**, Ibadurrahman, Fajri A. Rayhan. *Multi-hull Ship Resistance*. DTM Press, 2022. ISBN: 978-602-98412-9-9

INTELLECTUAL PROPERTY

Pushing the Utilization of Ocean Space on the Blue Energy System and Green Ship Technology

Creator: Kurniawan T. Waskito
Type: Video recording (recorded video work)
Registration No.: 000641773

JOURNAL ARTICLES

[1] Arrosyid W.A., Yanuar, Waskito K.T., Kurniawan A., Siringo Ringo E., Geraldi A., Gunawan, Zhou X., Kumarasamy S. Performance and survivability trade-off analysis of pitching-mode wave energy converters in intermediate water: A South Java case study. *Ocean Engineering*, *Vol. 370, 129233 (2026)*. Link

[2] Arrosyid W.A., Sari W.R., Waskito K.T., Yanuar. Evaluation and Optimization of Wave Energy Converters: A Systematic Perspective on Methods and Design Optimization Research Trend. *Engineering Journal*, *30(3), pp. 1–27 (2026)*. Link

[3] Waskito K.T., Tarigan L.G., Yanuar et al. Resistance and Propulsion Study of a 17,500-DWT Oil Tanker Using CFD and Slender-Ship Theory: Effects of Energy-Saving Device Fins. *J. Marine. Sci. Appl. (2026)*. Link

[4] Liu C., Jiang X., Li H., Sun S., Ren H., Riadi A., Waskito K.T. Experimental and numerical investigation of floating offshore substation dynamic responses with mooring line failures. *Ocean Engineering*, *Vol. 348 (2026)*. Link

[5] Arrosyid W.A., Waskito K.T., Geraldi A. et al. Hydrodynamic analysis of multi-cylinder floating-point-absorber wave energy converter. *J. Ocean Eng. Mar. Energy (2025)*. Link

[6] Geraldi A., Waskito K.T., Arrosyid W.A., Sari S.P., Yanuar. Estimation of Hydraulic Power Take-Off Parameters in a Floating Absorber Type Wave Energy Converter Using the Genetic Algorithm Method. *Eng. J.*, *29(7), pp. 1–15 (2025)*. Link

[7] Arrosyid W.A., Sari W.R., Waskito K.T., Yanuar, Utama I.K.A.P., Soesanto Q.M.B., Bramantya A., Nugroho B. Recent advancements in wave energy converter technologies: A comprehensive review on design and performance optimization. *Ocean Engineering 340(2) (2025) 122328*. Link

[8] Abimanyu R.F.S., Waskito K.T., Sari W.R., Ogata S. Impact of Reynolds Number on Flow Behavior in Spiral Pipe Profiles: A Review. *J. Adv. Res. Fluid Mech. Thermal Sci.*, *129(2), pp. 216–243 (2025)*. Link

[9] Sari W.R., Gunawan G., Waskito K.T. et al. Integrating Marine Renewable Energy with Green Hydrogen Production from Seawater: Feasibility and Future Prospects for Sustainable Energy Development in Indonesia. *J. Marine. Sci. Appl. (2025)*. Link

[10] Arrosyid W.A., Waskito K.T., Yanuar, Nasruddin, Sholahudin, Geraldi A., Zhao Y. Hydrodynamic performance and multi-objective optimization of multi-cylinder floating point absorber wave energy converter. *Ocean Engineering 317 (2025) 120040*. Link

[11] Waskito K.T., Siahaan J.A.C., Chuzain M.A.N., Yanuar, Pal S. Design Optimization of a Point Absorber and Hydraulic Power Take-Off Unit for Wave Energy Converter. *International Journal of Technology*, *15(5), pp. 1524–1538 (2024)*. Link

[12] Waskito K.T., Geraldi A., Ichi A.C., Yanuar, Rahardjo G.P., Al Ghifari I. Design of hydraulic power take-off systems unit parameters for multi-point absorbers wave energy converter. *Energy Reports*, *Vol. 11 (2024), pp. 115–127*. Link

[13] Waskito K.T., Yudho R.H., Yanuar Y., Rahardjo G.P. Evaluating Wave Potential and Assessing the Economic Viability of Wave Energy Converters in the South Java Seas. *Kapal*, *20(3), pp. 391–400 (2023)*. Link

[14] Waskito K.T., Sasa K., Chen C., Kitagawa Y., Lee S.-W. Comparative study of realistic ship motion simulation for optimal ship routing of a bulk carrier in rough seas. *Ocean Engineering*, *260 (2022) 111731*. Link

[15] Lee S.-W., Sasa K., Chen C., Waskito K.T., Cho I.-S. Novel safety evaluation technique for ships in offshore anchorage under rough seas conditions for optimal ship routing. *Ocean Engineering*, *253 (2022) 111323*. Link

[16] Waskito K.T., Yanuar. On the High-Performance Hydrodynamics Design of a Trimaran Fishing Vessel. *J. Adv. Res. Fluid Mech. Thermal Sci.*, *83(1), pp. 17–33 (2021)*. Link

[17] Waskito K.T., Kashiwagi M., Iwashita H., Hinatsu M. Prediction of nonlinear vertical bending moment using measured pressure distribution on ship hull. *Applied Ocean Research, Vol. 101 (2020) 102261.* Link

[18] Gunawan, Yanuar, Waskito K.T. Determination the optimum location for microbubble drag reduction method in self-propelled barge model: An experimental approach. *Energy Reports, 6 Suppl. 2 (2020), pp. 774–783.* Link

[19] Yanuar, Waskito K.T., Gunawan, Candra B.D., Perdana A.Y. Micro-bubble drag reduction by multi discrete hole plate on self-propelled barge ship model. *J. Adv. Res. Fluid Mech. Thermal Sci., 53(1), pp. 111–121 (2019).* Link

[20] Yanuar, Waskito K.T., Pratama S.Y., Candra B.D., Rahmat B.A. Comparison of microbubble and air layer injection with porous media for drag reduction on a self-propelled barge ship model. *J. Marine. Sci. Appl. 17, pp. 165–172 (2018).* Link

[21] Yanuar, Ibadurrahman, Waskito K.T., Karim S., Ichsan M. Interference resistance of pentamaran ship model with asymmetric outrigger configurations. *J. Marine. Sci. Appl. 16, pp. 42–47 (2017).* Link

[22] Yanuar, Waskito K.T., Mau S., Wulandari W., Sari S.P. Helical twisted effect of spiral pipe in generating swirl flow for coal slurries conveyance. *Jurnal Teknologi, 79(7-3) (2017).* Link

[23] Yanuar, Waskito K.T., Gunawan, Budiarso. Drag reduction and velocity profiles distribution of crude oil flow in spiral pipes. *International Review of Mechanical Engineering (IREME), 9(1), pp. 1–10 (2015).* Link

[24] Yanuar, Gunawan, Waskito K.T., Jamaluddin A. Experimental study resistances of asymmetrical pentamaran model with separation and staggered hull variation of inner side-hulls. *Int. Journal of Fluid Mechanics Research, 42(1), pp. 82–94 (2015).* Link

CONFERENCE PROCEEDINGS

[1] Geraldi A., Waskito K.T., Sari S.P., Yanuar. Performance Evaluation of a Wave Energy Converter Equipped with a Pneumatic-Driven Hydraulic Power Take-Off System. *Journal of Physics: Conference Series, IOP Publishing, Vol. 3186, 012032 (2026).* Link

[2] Gu N., Zhou X., Ren H., Waskito K.T. Study on wave loads of large multi-body floating offshore platform by model test. *IET Conference Proceedings, Vol. 2025, 46, pp. 245–250 (2026).* Link

[3] Waskito K.T., Yanuar. Development of wave-induced loads program for global ship hull strength. *AIP Conf. Proc. 2749, 020005 (2023).* Link

[4] Kitagawa Y., Tsukada Y., Waskito K.T. An experimental study on effects of propeller slipstream to hydrodynamic pressure on astern hull surface in waves (in Japanese). *JASNAOE Annual Autumn Meeting, Yokohama, Japan, 2021.*

[5] Kashiwagi M., Iwashita H., Waskito K.T., Hinatsu M. Prediction of wave loads with measured unsteady pressure distribution on ship hull surface. *35th International Workshop on Water Waves and Floating Bodies, Seoul, Korea, 2020.* Link

[6] Waskito K.T., Kashiwagi M., Iwashita H., Hinatsu M. Prediction of nonlinear wave loads using measured pressure distribution on ship hull. *International Symposium on Novel Computational and Experimental Methods for Complicated Fluid-Structure Interactions, Fukuoka, Japan, 2020.*

[7] Waskito K.T., Kashiwagi M. Improvement of wave loads estimation from spatial pressure distribution on ship hull. *ASME OMAE2019, Glasgow, UK, 2019.* Link

[8] Yanuar, Waskito K.T. Experimental Study of Total Hull Resistance of Pentamaran Ship Model with Varying Configuration of Outer Side Hull. *Procedia Engineering, 10th MARTEC 2016, Dhaka, Bangladesh, 2017.* Link

AWARDS & SCHOLARSHIPS

- Outreach scholarship for the engineering specialist forum at OMAE 2019, Glasgow, UK
- Doctoral scholarship from the Japan International Cooperation Agency (JICA) – Innovative Asia, Osaka University, Japan. October 2017 – September 2020
- Best Presentation Award, 9th International Meeting on Advances in Thermofluids (IMAT2017), Johor, Malaysia
- Master's scholarship from the Indonesian Directorate General of Higher Education (DITJEN-DIKTI), fast-track programme, Universitas Indonesia. September 2012 – August 2014
- Outstanding Student, Naval Architecture Academic Award, 2011, Faculty of Engineering, Universitas Indonesia
- Finalist, National Student Scientific Week (PIMNAS) XXIV, PKM-GT, 2011, DITJEN-DIKTI, UNHAS, Makassar, Indonesia

Depok, 29 April 2026
Kurniawan Teguh Waskito, Ph.D.